Tsukuyomi Watch

The Samurai’s Celestial Guide

Plan the night.
Read the sky.
Inspect the image.

Weather + cosmic events + conservative local AI in one Japanese-inspired observing workflow.

tsukuyomi-watch.vercel.app



LIVE DEMO + VERIFIED TEST EVIDENCE

A clear night is a decision problem

A beginner must combine fragmented information before an observing trip.

01 / WHERE

Will the sky be usable?

Clouds, humidity and visibility change by place and time. A map alone cannot rank an observing night.

02 / WHEN

What can I observe?

Moon phases, launches, eclipses, meteor windows and visible ISS segments come from different sources.

03 / WHAT

Can I trust the image?

Noise and blur hide faint structure. Cleanup must stay inspectable and avoid invented object identities.

Primary users: student astronomy clubs, first-time observers and hobby astrophotographers.

One observing journey, four connected views

The core PS-002 modules are implemented; the interface also includes optional bonus tools.

01 / WATCHTOWER

Choose a place

Search the scroll map.
Compare nighttime forecasts.

→

02 / CELESTIAL SCROLL

Choose a moment

Filter cosmic events.
Read the lunar calendar.

→

03 / SPIRIT LENS

Restore carefully

Run local AI denoising.
Inspect object candidates.

→

04 / DOJO / VIEWER

Inspect and record

Compare, zoom and export.
Log an observation.

BONUS: archival SkyView import • CSV signal screening • local Honor Points

The contribution is the observing workflow

We combine established science and AI tools with explicit provenance and a distinctive interface.

CONNECTED PLANNING

Place → time → image

Weather ranking, lunar navigation, event details, alerts and inspection are available in one application.

INSPECTABLE AI

Local and bounded

A browser worker runs DnCNN. Conservative edits, object overlays and downloadable measurements support review.

CULTURAL INTERACTION

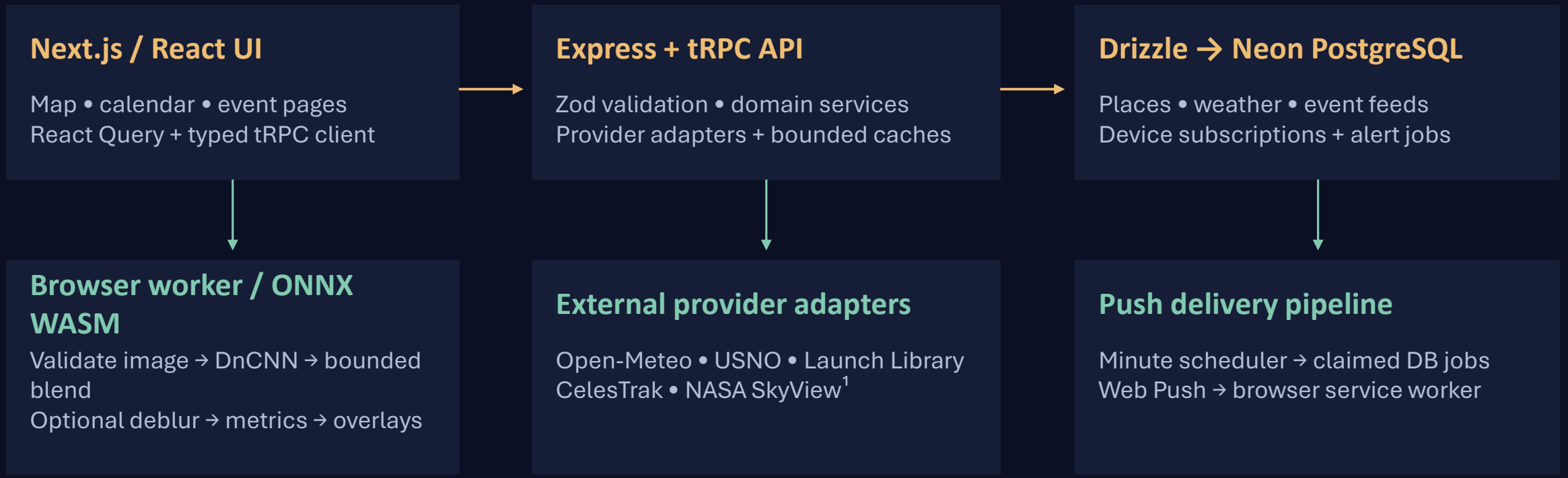
A celestial scroll

Ink Wash, Torii, Sakura and Katana palettes; light/dark mode; moon cues; English meanings for Japanese labels.

Originality claim: integration and experience. DnCNN, SGP4 and astronomical algorithms are established methods.

Typed boundaries, separated responsibilities

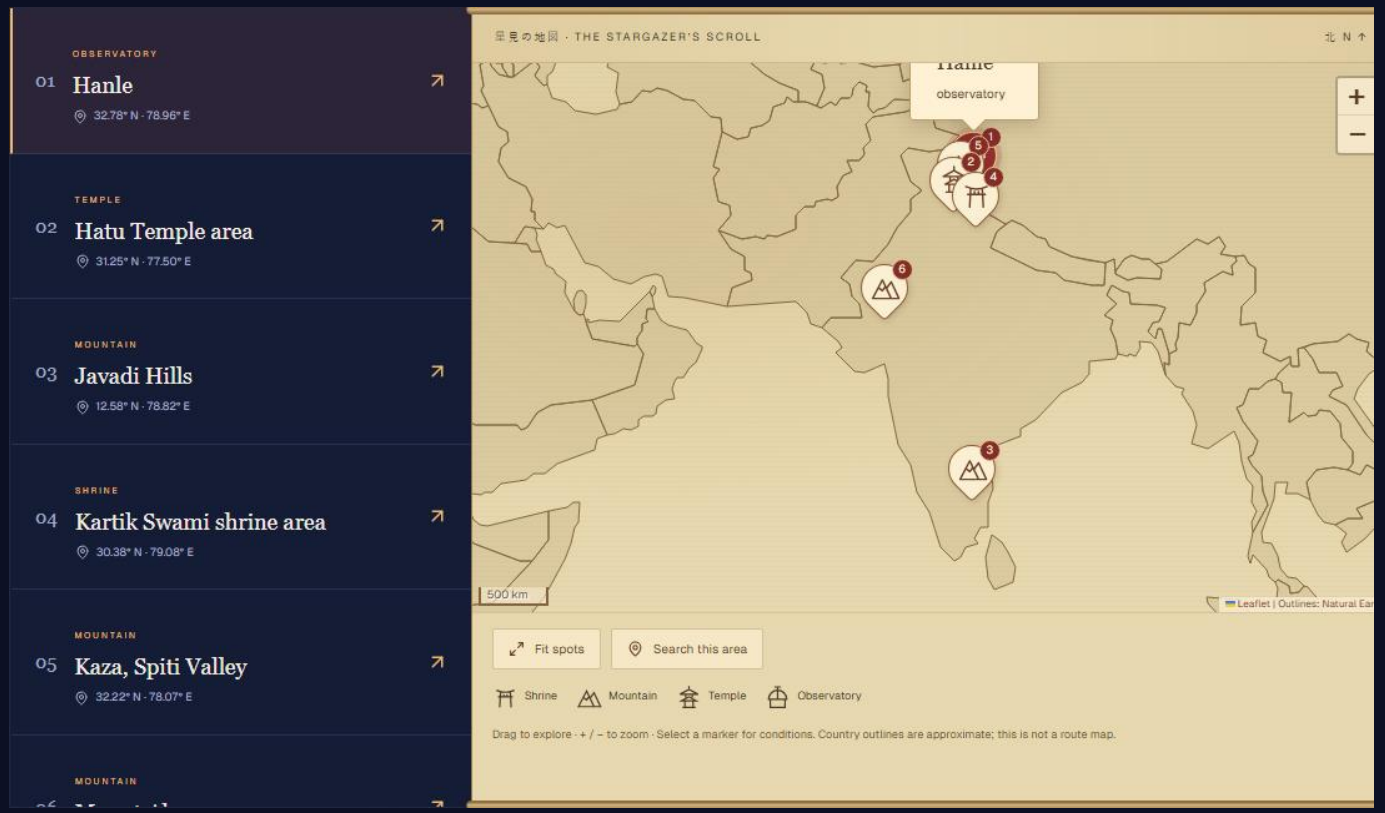
Image pixels stay in the browser; planning data and alert jobs use the backend.



[1] SkyView uses a fixed-host Next.js server route. Vercel frontend • Render API • Neon database.

A scroll map backed by nighttime forecasts

Search a place or map area, select a marker and compare the next available nighttime conditions.



- Open-Meteo cloud cover, humidity and visibility.
- Night Honor Score: transparent weather-only ranking.
- Shrine, mountain, temple and observatory markers.
- Six curated Indian destinations today.

Time-aware events, with uncertainty visible

A Japanese Kyūreki month view includes lunar leap months and Gregorian dates.



- Moon phases, launches, eclipses and meteor windows.
- Visible ISS segments use fresh orbital elements.
- Internal event details and calendar export.
- Moonrise reminders and optional browser push.

Conservative cleanup with an audit trail

DnCNN denoises a preview locally; users retain control over every result.

VALIDATE

PNG / JPEG / WebP
12 MB • 24 MP cap

→

NORMALIZE

RGB preview
Longest edge
≤768px

→

RESTORE

Verified DnCNN
256px tiled
inference

→

CONSTRAIN

Bounded luminance
Protect source
regions

→

INSPECT

Compare + zoom
PNG / JSON export

OPTIONAL DEBLUR

Six Richardson–Lucy iterations

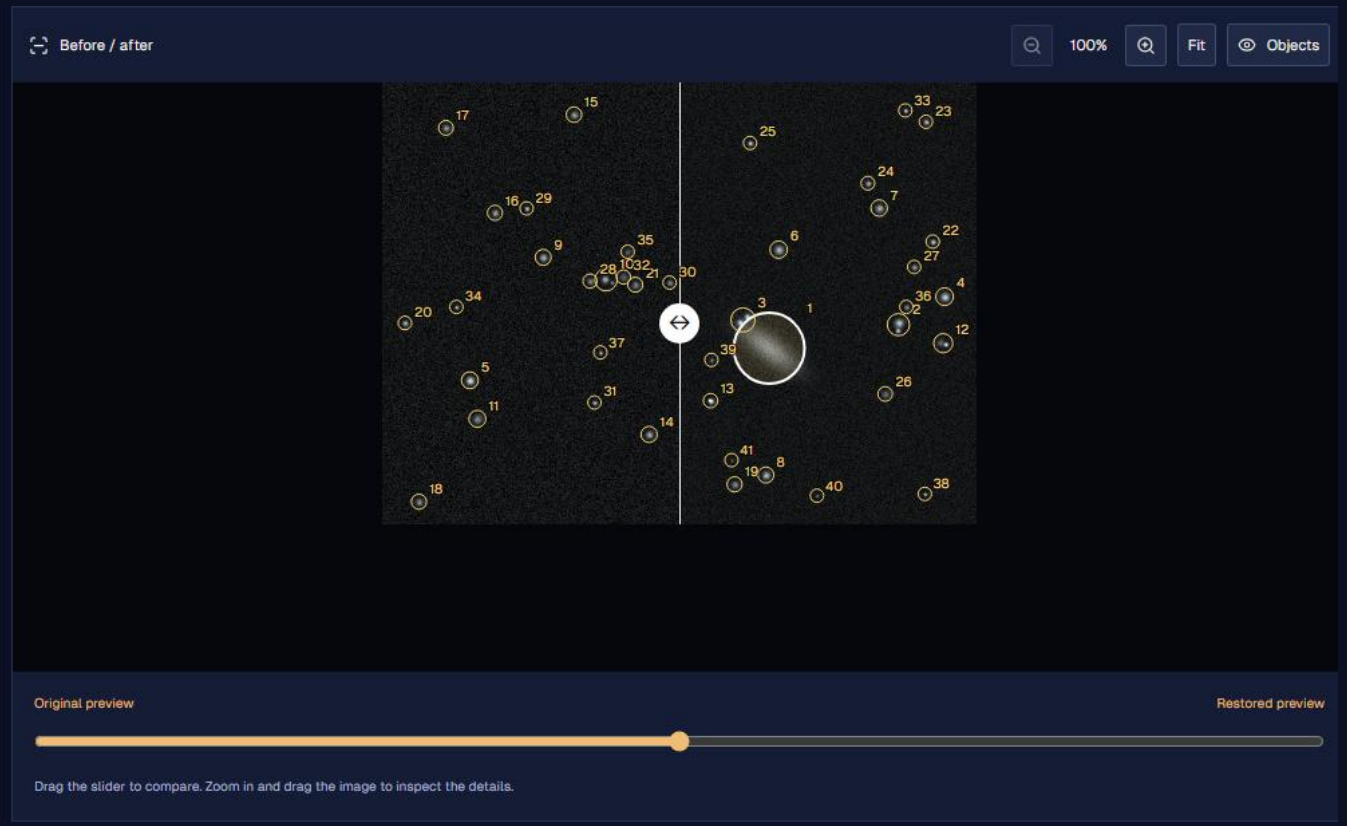
User-selected Gaussian PSF; wrong settings can cause ringing.

SCIENTIFIC LIMIT

Bounded edits reduce risk. They cannot guarantee that every faint feature remains unchanged.

Compare pixels, not promises

Synthetic demo fixture • 444 × 330 pixels • Gentle mode • one production browser run



15.8 s
This browser run

41
Object candidates

8.6 → 6.5
Estimated noise

6 / 255
Max channel edit

SSIM 0.975 compares output with noisy input.
It does not prove clean-image accuracy.

28 automated tests pass across 8 files

Fresh audit run: 30 October 2026 IST • 28 passed • 0 failed • no coverage percentage claimed

AREA	TESTS	EDGE CASE EVIDENCE
Weather + distance	4	Score bounds; missing night data; date-line crossing
Moon + reminders	3	Polar null rise/set; UTC and observer timezone
Event providers / export	4	Leap February; source IDs; ICS injection / Unicode
Eclipses / lunar / ISS	5	Known eclipse; leap lunar month; invalid orbital data
Push security	2	SSRF endpoint rejection; invalid scheduler credentials
Lens analysis / deblur	6	RGB/alpha bounds; flat sky; image limits; blur fixture
Signals + Honor	4	Zero scatter; bad CSV; duplicate / future observations

Production builds: Next.js frontend PASS • tsup API PASS

A useful student-scale platform, live today

Deployed on Vercel + Render + Neon, with public APIs and browser-side inference.

USER VALUE

Fewer disconnected steps

Compare places, plan around events and inspect a restored image without switching between separate tools.

PRACTICAL DELIVERY

A working public demo

Local AI avoids an inference server. Provider adapters, caching and bounded inputs keep the prototype feasible.

VALIDATE NEXT

Measure real benefit

Pilot with an astronomy club. Measure planning time, successful observing trips and faint-source preservation.

No fabricated adoption or accuracy claims. Free hosting can sleep; third-party APIs can be unavailable.

Ready for a controlled hackathon demonstration

The implementation is substantial; a maximum score or production-grade reliability is not established.

CRITERION	MARKS	ASSESSMENT / PREPARATION
Innovation	15	Strong integration story; distinguish it from new research.
System design / code	20	Clear layers and typed contracts; rehearse a request trace.
Problem understanding	15	All four core modules; explain estimates and source limits.
Test cases	20	28 passing tests; CI, device push and broader E2E remain.
Impact / feasibility	15	Live prototype; club pilot and scale evidence remain.
Presentation	15	Editable deck, demo script and screenshot fallback ready.

Before judging: warm the API, rehearse the demo, keep local image/CSV inputs and explain push honestly.

A five-minute demo with a reliable fallback

Open the public URL before the pitch and keep the screenshot deck available.

0:00–0:30	**Problem + promise**	Choose a place, a moment and a trustworthy image workflow.
0:30–1:20	**Watchtower + Scroll**	Search / select a spot, inspect conditions, open an event.
1:20–2:30	**Spirit Lens**	Use the labelled demo, restore, drag slider, zoom and inspect.
2:30–3:10	**Bonus tools**	Show archive import, a synthetic CSV result and Honor journal.
3:10–4:20	**Architecture + tests**	Trace one request and show the 28-test evidence.
4:20–5:00	**Impact + limits**	Explain the pilot plan, provider uncertainty and push setup.

Know where every responsibility lives

Use these paths to move from a judge's question to the implementation.

UI + feature pages	apps/web/app; apps/web/components
Typed API contracts	packages/trpc/server/routes; apps/api/src/server.ts
Weather / ranking / moon	packages/services/watchtower
Events / Kyūreki / ISS	packages/services/celestial
Persistent data / migrations	packages/database/models; packages/database/drizzle
AI / object analysis / deblur	apps/web/lib/spirit-lens
Signals + local Honor	apps/web/lib/transient-analysis.ts; honor-journal.ts
Push delivery / scheduler	packages/services/alerts; ops/push-cron.mjs

Explain assumptions before giving a score

Night Honor Score is a weather heuristic, not a scientific observability guarantee.

$$\text{Score} = \text{round}\{100 \times [40(1 - C/100) + 20\ \min(V/20, 1) + 15(1 - H/100)] / 75\}$$

C = cloud cover % V = visibility in km H = relative humidity %

- Bounds: 0–100; visibility gets full credit at 20 km.
- Forecast uses the next future provider-night hour; null values are rejected.
- Moon illumination and light pollution are currently excluded.
- Meteor windows are approximate; eclipse maxima are global; ISS passes are predictions.

What the evidence proves — and what remains

Passing tests establish selected behaviors; they do not establish exhaustive coverage.

VERIFIED IN THIS AUDIT

Repeatable evidence

28 / 28 automated test cases pass.
Frontend and API builds pass.
Production map and calendar render.
Browser AI completes with exported metrics.

PRIORITY NEXT

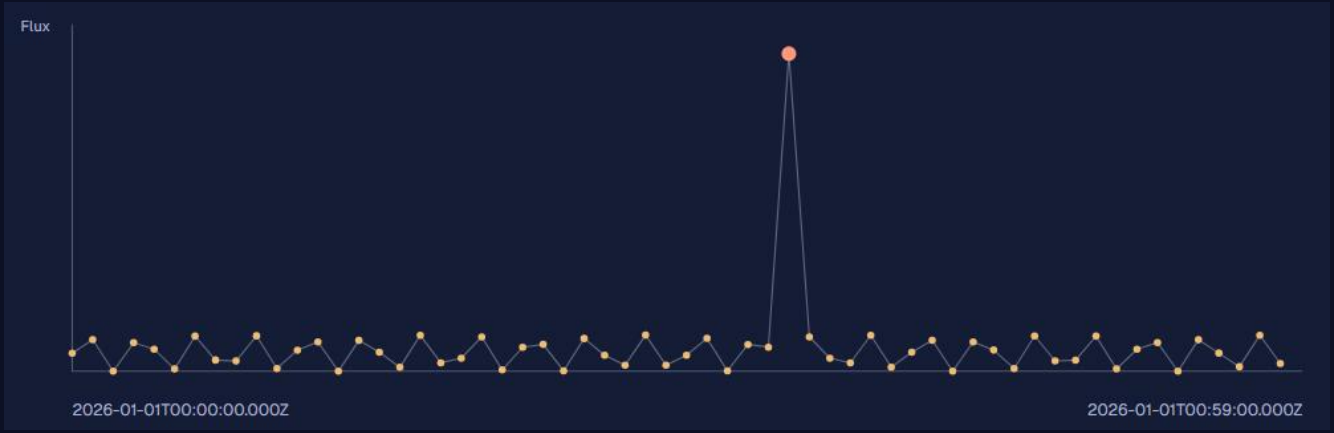
Close the evidence gaps

CI for tests and builds.
Permission / offline / mobile E2E.
Physical closed-app push receipt.
Real astronomy and load benchmarks.

Earlier PostgreSQL ownership integration is separate evidence; it was not rerun in this audit.

Useful extras, with explicit boundaries

These features are implemented as prototype tools, not confirmed scientific discovery services.



Synthetic light-curve screening output

- SkyView: on-demand archival survey JPEGs, not live telescope video.
- Signals: CSV median / MAD screening, not confirmed transient discovery.
- Honor: 25 points per unique observation, stored on this device.
- Push: persistent jobs exist; closed-app delivery needs an active scheduler.

NEXT: more destinations • calibrated validation • authenticated sync • reliable scheduling